Chapter 3

Relevance of Particle Transport in Surface Deposition and Cleaning

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Chapter Outline

3.1 INTRODUCTION

Through numerous natural phenomena as well as human activities, we have ample experience about foreign objects in various fluid media and their interactions. To name a few that occur naturally are sandstorms, precipitation, cloud formation, and botanical reproduction. In our daily economic life activities, such as pest control through spraying, dust collection, mining and manufacturing processes, and transportation and pollution control are just a tiny fraction of examples that involve interaction of droplets, vapors, particles, and bubbles with various fluid media. Physically speaking, we live in a world of tremendous systems of fluid–particle interactions, as first delineated by Soo in 1965.[1] Researches on this subject and its related phenomena have progressed significantly since the second half of the last century.[2–4] In this chapter, we are focusing primarily on the recent progress in gas-particle flows, in general, and their relevance to fine particle surface deposition/cleaning, in particular.

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The characteristics of gas-particle flows vary widely with respect to the geometric and material properties of the particle.[4] In solid fuel combustion, pneumatic conveying, fluidization, gas-particle separation, particle deposition on semiconductor wafers[5], and other processes, gas-particle flows are commonly witnessed. Likewise, gas-particle flows play important roles in various natural phenomena such as dry deposition of particulate matters on natural surfaces,[6] dispersion of natural allergens, volcanic ashes, cosmic dusts, etc. Surface contamination due to the presence of fine particles has caused enormous yield loss and degradation of reliability in microelectronics manufacturing[7] and material processing,[8] damages in agricultural and forestry products[9] and art works.[10] In general, deposition of atmospheric contaminants onto solid surfaces is a dynamic process. This process is carried out in three steps as follows: (1) particles are transported aerodynamically to the proximity of the boundary layer surrounding the solid surfaces, (2) particles are transported within the boundary layer and toward the gas–solid interfaces, and (3) particles come physically in contact with the solid surfaces.[11]

In this chapter, descriptions of the important mechanisms that are involved in this process are provided. In Section 3.2, particle transport in gas-particle flows related to surface deposition is described. At the particle–solid interfaces, the primary adhesion mechanisms are also detailed and discussed in Section 3.2. Subsequently, Section 3.3 presents the various mechanisms of the aerodynamic and boundary layer transport of particles. The relevance of the particle transport theories is given. In Sections 3.4–3.6, three mechanisms, namely thermophoresis, electrostatics and dielectrophoresis, and their implications in particle deposition are delineated. Section 3.7 is devoted to a unique particle deposition problem, namely abrasive erosion and its relevance in surface cleaning. The recent progress in numerical and experimental studies on particle transport phenomena described in the aforementioned sections is summarized.

3.2 PARTICLE–SOLID SURFACE INTERACTIONS

For understanding the various phenomena involved in surface contamination by particles to developing means for monitoring and removing the contaminants, it is necessary to learn and apply the mechanics of gas-particle flows. Fuchs[12] and Friedlander[13], in their widely cited books, have compiled and provided the most up-to-date knowledge on the motion of aerosols (i.e., solid or liquid particles) in natural or man-made environments in the 1960s and 1970s, respectively. Hinds summarized further progress in aerosol technology in his well-received textbook.[14] Most recently, Fan and Zhu have given a thorough description of the basic principles and fundamental phenomena associated with gas–solid flows.[4] In this chapter, from the application perspectives of surface deposition and cleaning, we focus our discussion on the important mechanisms involved in the context of gas-particle flows. For more rigorous theoretical derivations, the readers are encouraged to refer to the references cited in this chapter.

When a solid particle physically contacts either another solid particle or a solid surface, the forces between these solid objects are primarily attractive in nature and result in adhesion of particles to each other or to the solid surface. The finer the size of the particles, the adhesion forces involved become more significant and particles become more difficult to remove.[15] Particle agglomeration due to adhesion could change the particle size distribution and other physical properties in the gas–solid system. Consequently, this effect would alter the effectiveness of particle removal. From the perspectives of surface deposition of particles, the most affected industry is the manufacturing of microelectronics.[16] Therefore, the entire clean room technology was mostly built with the aim of controlling the deposition of fine particles to reduce yield losses during the manufacturing of integrated circuits.

After a physical contact between particles and a solid surface is established, it is considered that van der Waals and electrostatic forces are the predominant attractive forces between the adhering particles. Other interfacial forces such as liquid bridges, double-layer repulsion and chemical bonds may play important roles under specific circumstances.[15,16] Penney and Klingler[17] have reported that, by measuring the contact potential differences of adjoining dust particles, the combination of van der Waals and electrostatic forces is normally an order of magnitude greater than the adhesion force by mechanical means. Figure 3.1 shows a schematic of adhesion by the van der Waals forces. For a spherical particle depositing on a planar surface, the resulting van der Waals forces are expressed as

$$F_{\mathrm{vdw}} = A\frac{d_{\mathrm{p}}}{z_0^2} \tag{3.1}$$

where F_{vdW} is the van der Waals forces, A is the Hamaker constant, d_{p} is the particle diameter, and z_0 is the clearance between the particle and the surface.[14,16]

For the same configuration, as shown in Figure 3.1, the difference in the work functions of the two materials results in a contact potential, ϕ_{c}, between

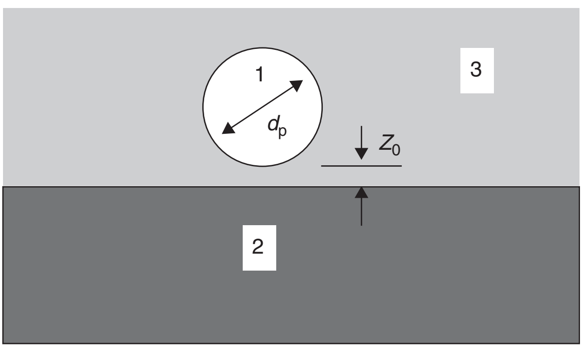

FIGURE 3.1 van der Waals interaction: sphere-planar surface.[15]

the particle and the surface. The force due to the electrostatic double-layer is given[15] as

$$F_e = \pi \varepsilon_0 \frac{d_p}{2} \frac{\phi_c^2}{z_0} \tag{3.2}$$

where F_e is the force due to the electrostatic double-layer and ε_0 is the dielectric constant of vacuum. With the common presence of humidity in the environment, it is not unusual to have some moisture adsorbed on the surface of most materials. At the interface where two different materials contact, a liquid bridge (or film) forms and results in an attractive force due to the surface tension of the liquid, as shown in Figure 3.2. For an ambient relative humidity above 90%, this attractive force is given as

$$F_s = 2\pi \gamma d_p \tag{3.3}$$

where F_s is the adhesion force due to the surface tension, γ, of the liquid bridge. The major adhesion forces mentioned above are all proportional to the particle diameter, d_p. In contrast, to remove the particle by mechanical means, we need to employ either a centrifugal force that is proportional to d_p^3 or an air current force that is proportional to d_p^2. Therefore, much greater force is needed to remove deposited micrometer and sub-micrometer particles. Based on measurements of hard materials and clean surfaces at 298 K, an empirical formula for the overall adhesion force, F_{ad}, is given as[14]:

$$F_{ad} \cong 150 d_p [0.5 + 0.0045(\%\mathrm{RH})] \tag{3.4}$$

For spherical particles of unit density, a comparison among the adhesion force, calculated using Eq. (3.4), and the corresponding gravitational and aerodynamic drag forces is made in Table 3.1. This example illustrates the significance of adhesion force and the difficulty in dislodging the deposited particles from the surface, especially for fine particles.

Even though we can calculate the primary mechanisms of particle–solid surface adhesion based on the equations above, in reality, however, the distance of separation, z_0, is not measurable and an assumed range from 0.4 to 1.0 nm is used. Therefore, this introduces a certain degree of uncertainty into both the measured and predicted data. In addition, approximations are made to define the roughness and asperities of the adhering bodies, as illustrated in Figure 3.3. Further research to improve the accuracy of adhesion forces, experimentally and computationally, is rightly justified.

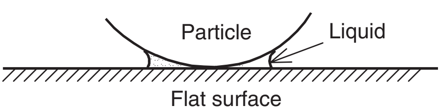

FIGURE 3.2 Adhesion force due to a liquid film.

TABLE 3.1 **Comparison of Adhesion, Gravity, and Air Current Forces on Spherical Unit-density Particles**[14]

Diameter (μm)	Force (N)		
	Adhesion[a]	Gravity	Air Current (at 1 m/s)
0.1	10^{-8}	5×10^{-18}	2×10^{-10}
1.0	10^{-7}	5×10^{-15}	2×10^{-9}
10	10^{-6}	5×10^{-12}	3×10^{-8}
100	10^{-5}	5×10^{-9}	6×10^{-7}

[a]*Calculated by Eq. (3.4) for 50% RH.*

Numerous ways, dry and wet, have been proposed and investigated to remove particles from surfaces. To reduce contamination, researchers and practitioners, though not always in agreement, prefer dry approaches, as reported by Cooper et al.[7] and Ranade.[16] It is known that blowing dry air or nitrogen jet is not effective to remove particles smaller than 10 μm in diameter. Cooper et al.[7] applied an electric field between an electrostatic film cleaner and the particle-deposited product to remove particles by electrostatic force, as shown in

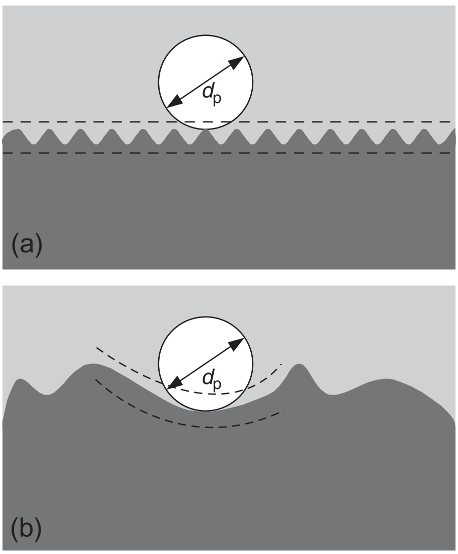

FIGURE 3.3 Effect of surface roughness on particle deposition.[15] (a) Asperities smaller than particle size. (b) Asperities larger than particle size.

Figure 3.4. Their success was not universal with respect to all the particles they had tested, as summarized in Table 3.2.[7] Ranade[16] reviewed many removal schemes and processes, both dry and wet, and no preferred technique was recommended due to the complexity involved in the phenomena of surface deposition of particles.

3.3 DRY DEPOSITION

Dry deposition of particulate matter on natural systems such as vegetated canopies and natural water bodies has been studies for several decades.[6,9,11] For a wide variety of biotic and natural surfaces, Sehmel gave a detailed review of published dry particle and gas deposition velocities that ranged over three and four orders of magnitude, respectively.[6] The factors that may affect dry deposition of particles and gas species are summarized in Table 3.3.[6] The mechanisms for dry deposition could be inertial impaction,[18–22] interception,[23,24] diffusion,[5,25–27] sedimentation,[5,25,26] and electrostatic precipitation and thermophoresis.[28] For surface deposition of particles in many industrial processes, inertial impaction and interception are the dominant forces encountered in particle deposition on filters, work benches and in particle sampling of high purity gases for semiconductor manufacture. On the other hand, for finer particles (such as micrometer and sub-micrometer particles), Brownian diffusion emerges as one of the primary mechanisms for deposition of particles on surfaces.

Conventionally, to study particle deposition on a surface, many researchers favor a rotating disk in a stagnation flow setting.[8,23] Ramarao and Tien[23] proposed a procedure to study the combined effect of inertial impaction,

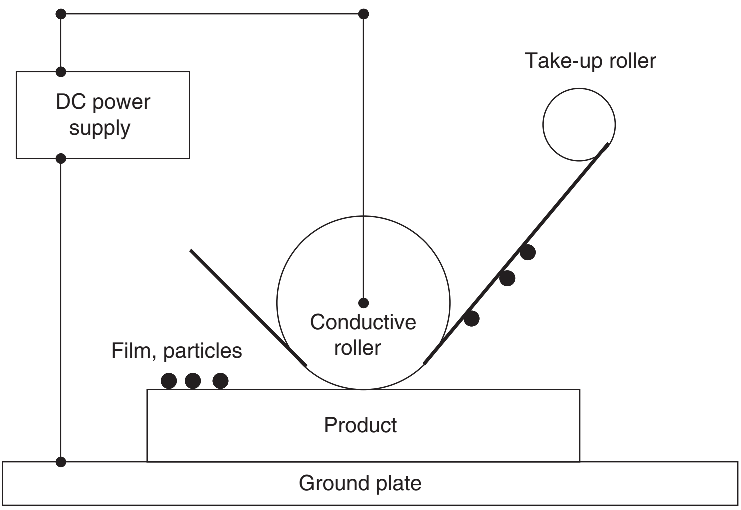

FIGURE 3.4 Schematic diagram of the electrostatic film cleaner.[7]

TABLE 3.2 Summary of Measured Removal Efficiencies, Using an Electrostatic Film Cleaner[7]

Particle Diameter (μm)	Particle Type	Film Used	Removal Efficiency (%)	Field (MV/cm)	Voltage (kV)
Conductive surfaces—smooth (specular)					
Metal film					
1.1	Latex	Polyimide	>46–98	2.8	–
0.5	Latex	Polyimide	>95	7.0	–
0.3	SiO_2	Polyimide	>95	8.4	–
Silicon wafer					
6	Ni	Laminate	80–100	–	4
2	SiO_2	Laminate	>95	–	6
0.5	SiO_2	Laminate	>80	–	6
Conductive surfaces—patterned or rough					
Semiconductor chips post-metallization					
2–5	Ni	Laminate	60–90	–	14
2	SiO_2	Laminate	>50	–	14
Insulator surfaces—smooth (specular)					
SiO_2 (0.5 μm) silicon wafer					
2–5	Ni	Laminate	98	–	4
2	SiO_2	Laminate	97	–	3
1	Latex	Laminate	~60	–	3
Ceramic substrate					
12	SiO_2	Laminate	96–100	–	10
6	Ni	Laminate	96–100	–	10
Glass plate. 1.5 mm thick					
6	Ni	Laminate	93	–	8
2	SiO_2	Laminate		–	8
0.5	SiO_2	Laminate		–	8
Insulator surfaces—patterned or rough					
Multilayer ceramic with metal pads					
12	SiO_2	Laminate	97	–	3–4
6	Ni	Laminate	95–99	–	3–4

TABLE 3.3 Some Factors that Influence Dry Deposition of Particles[6]

Micrometeorology Variables	Deposition Material: Particles	Deposition Material: Gases	Surface Variables
Aerodynamic roughness	Agglomeration	Chemical activity	Adhesion
—Mass transfer	Diameter	Diffusion	**Biotic surfaces**
(a) Particles	Density	—Brownian	Canopy growth
(b) Gases	Diffusion	—Eddy	—Dormant
—Heat transfer	—Brownian	Partial pressure in equilibrium with surface solubility	—Expanding
—Momentum transfer	—Eddy equal to		Senescence
Atmospheric stability	(a) Particle		Canopy structure
Diffusion, effect of	(b) Momentum		—Area density
—Canopy	(c) Heat		—Bark
—Diurnal variation	—Effect of canopy on diffusiophoresis		—Bole
—Fetch	Electrostatic effects		—Leaves
Flow separation	—Attraction		—Porosity
—Above canopy	—Repulsion		—Reproductive structure
—Below canopy	Gravitational settling		—Soils
Friction velocity	Hygroscopicity		—Stem
Inversion layer			—Type
Pollutant concentration	Impaction		Electrostatic properties
Relative humidity	Interception		Leaf-vegetation
Seasonal variation	Momentum		—Boundary layer
Solar radiation	Physical properties		—Change at high winds
Surface heating	Re-suspension		—Flutter

TABLE 3.3 Some Factors that Influence Dry Deposition of Particles—Cont'd

Micrometeorology Variables	Deposition Material		Surface Variables
	Particles	**Gases**	
Temperature	Shape		—Stomatal resistant
Terrain	Size		**Non-biotic surfaces**
—Uniform	Solubility		pH effects on
—Non-uniform	Thermophoresis		—Reaction
Turbulence			—Solubility
Zero-plane displacements			Pollutant penetration and distribution in canopy
—Mass transfer			
(a) Particles			Prior deposition loading
(b) Gases			Water
—Heat transfer			
—Momentum transfer			

sedimentation and interception on particle deposition in a two-dimensional stagnation flow field. For particle size ranges from 1 to 100 μm in diameter, they found that the deposition flux, J, was greater by taking the boundary layer effect (denoted as J_{BL}) into account than by excluding it (denoted as J_{ID}). A correction factor, $F = J_{ID}/J_{BL}$, for the effect of the boundary layer on particle deposition under a range of the flow field constant, α, is obtained, as shown in Figure 3.5. Thus, a laminar flow is usually maintained in microelectronics manufacturing in cleanrooms. Usually finer particles whose diameter ranges between 0.01 and 10 μm are deposited on semiconductor wafers and diffusion is known to be the predominant deposition mechanism. By using the analogy with heat and mass transfer, Liu and Ahn[5] deduced the particle deposition velocity for three laminar flow settings typically used in the manufacturing environment for semiconductor wafers, as shown in Figure 3.6. For a horizontal wafer in a vertical airflow configuration, as shown in Figure 3.6(a), the mass transfer is given as

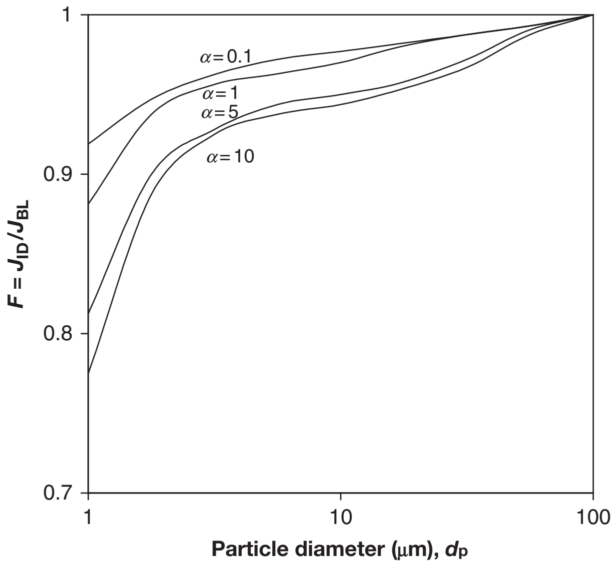

FIGURE 3.5 Flux correction factor, F, as a function of particle diameter for various values of α, flow field constant.[23]

$$\overline{\mathrm{Sh}} = 1.08\mathrm{Sc}^{1/3}\mathrm{Re}^{1/2} \tag{3.5}$$

and

$$\mathrm{Sh}_0 = 0.834\mathrm{Sc}^{1/3}\mathrm{Re}^{1/2} \tag{3.6}$$

where $\mathrm{Sh} = KD_\mathrm{w}/D$ is the mean Sherwood number, K is the mass transfer coefficient, D_w is the wafer diameter, $D = kTC/(3\pi\mu d_\mathrm{p})$ is the diffusion coefficient, k is the Boltzmann constant, T is the absolute temperature, C is the Cunningham slip correction factor, μ is the molecular viscosity of the fluid, $\mathrm{Sc} = v/D$ is the Schmidt number, $\mathrm{Re} = UD_\mathrm{w}/v$ is the Reynolds number, U is the airflow velocity, and v is the kinematic viscosity.

In Eqs. (3.5) and (3.6), the symbol $\overline{\mathrm{Sh}}$ denotes the mean Sherwood number based on the mean mass transfer coefficient, $\overline{K}$, for the entire wafer; and Sh_0 denotes the Sherwood number at the stagnation point. Note the definition of $K = j/N$, where j is the particle deposition flux in number of particles per unit surface area of the wafer per unit time and N is the number concentration of particles in the free stream. Therefore, K is also the deposition velocity of the particles. For the airflow configuration depicted in Figure 3.6(a), the overall deposition velocity, V, becomes

$$V = K + \rho_\mathrm{p} g C d_\mathrm{p}^2 / 18\mu \tag{3.7}$$

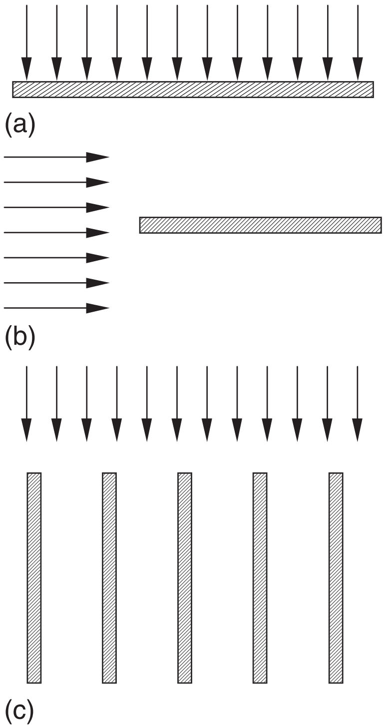

FIGURE 3.6 Schematic diagram showing three types of airflow in semiconductor manufacturing.[5] (a) A horizontal wafer in a vertical laminar flow cleanroom. (b) A horizontal wafer in a horizontal-flow clean hood. (c) Vertical wafers in a wafer carrier.

where the second term on the right-hand side of the equation is the sedimentation velocity, ρ_p is the material density of the particles, and g is the gravitational acceleration.

For a horizontal wafer in a horizontal airflow configuration, as illustrated in Figure 3.6(b), the mean and the local mass transfers are given in Eqs. (3.8) and (3.9) and expressed as

$$\overline{\mathrm{Sh}_x} = 0.664\mathrm{Sc}^{1/3}\mathrm{Re}_x^{1/2} \tag{3.8}$$

and

$$\mathrm{Sh}_x = 0.332\mathrm{Sc}^{1/3}\mathrm{Re}_x^{1/2} \tag{3.9}$$

where $\overline{\mathrm{Sh}_x} = \overline{K}x/D$, $\mathrm{Sh}_x = Kx/D$ and $\mathrm{Re}_x = Ux/v$ are all based on the distance, x, measured from the leading edge of the wafer and the air velocity, U. By performing an integration of K over the entire surface area of the wafer, we have

$$\overline{\mathrm{Sh_D}} = 0.739\mathrm{Sc}^{1/3}\mathrm{Re}_\mathrm{D}^{1/2} \tag{3.10}$$

where $\overline{\mathrm{Sh}_D} = \overline{K}D_w/D$, $\mathrm{Sh_D} = KD_\mathrm{w}/D$ and $\mathrm{Re_D} = UD_\mathrm{w}/\nu$ are all based on the wafer diameter, $\mathrm{D_w}$.

As shown in Figure 3.6(c), the third configuration is an array of vertical wafers in a wafer carrier. The mean mass transfer coefficient, $\overline{K}$, is given by

$$\overline{K} = (2h/\pi R_\mathrm{W})\overline{U}f_\mathrm{d} \tag{3.11}$$

where h is the half width of the channel formed by two neighboring wafers, R_w is the radius of the wafers, $\overline{U}$ is the mean air velocity in the channel, and f_d is the fractional particle deposition and is expressed as

$$f_\mathrm{d} = 97.27\varsigma^{0.6654} \tag{3.12}$$

where $\varsigma = DD_\mathrm{w}/h^2\overline{U}$ is a dimensionless parameter. For the first configuration, as shown in Figure 3.6(a), Liu and Ahn's prediction on particle deposition velocity was in good agreement with the measured data reported by Pui, Ye and Liu using ammonium fluorescein aerosols,[26] as depicted in Figure 3.7.

By taking convection, diffusion, sedimentation, electrical forces, and thermophoresis into account, a numerical simulation of particle deposition on a typical semiconductor wafer in a cleanroom using two different modeling approaches was reported by Peterson et al.[28] The predicted particle deposition velocity and flux showed a good agreement between the two approaches (i.e. a boundary layer approach and a full numerical simulation) they had adopted. For particle sizes between 0.02 and 0.7 μm at a surface temperature of 330 K, they found that the effect of thermophoresis could be effectively utilized to prevent such particles from depositing on a surface.[28]

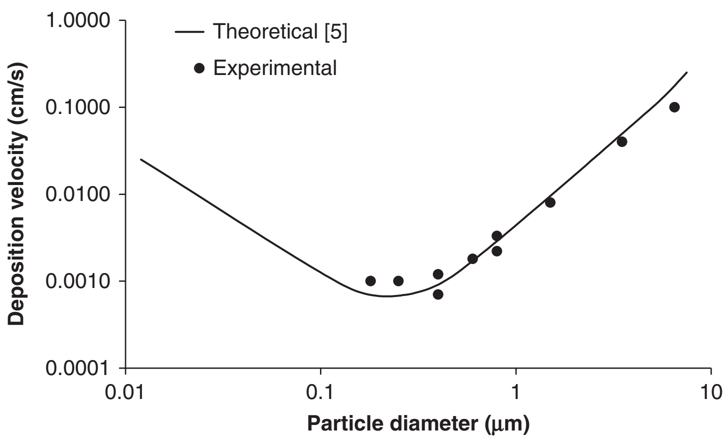

FIGURE 3.7 Comparison of the final deposition data from the theory of Liu and Ahn[5] with experimental data of[26] for 3.8-cm diameter wafer and for a free stream velocity of 20 cm/s.

LIVE GRAPH
Click here to view

3.4 THERMOPHORESIS AND ITS RELEVANCE IN SURFACE CLEANING

The phenomenon of particle repulsion from hot surfaces was first observed in the 1880s. However, the theoretical interpretation of the mechanism is still far from complete due to the complexity of the physical processes involved and the new findings obtained recently.[12,29–34] The radiometric forces acting on particles due either to a temperature gradient in the gas medium or to non-uniform radiation cause this phenomenon. The radiometric motion of particles by temperature gradient is called thermophoresis while the motion of particles by non-uniform radiation is called photophoresis.[2,12,35,36] If, instead a concentration exists and causes particle movement, this phenomenon is then called diffusophoresis.[37,38] Thermophoresis has been studied by numerous researchers to understand its effect on particle deposition on a cold surface,[10,39,40] or on particle removal from a heated surface, or in a thermal precipitator.[14,41,42]

In the air, for spherical particles ranging between 0.05 and 1 μm in diameter, the corresponding Schmidt numbers, $Sc = \nu/D$, are from 6×10^3 to 5×10^5. Therefore, Brownian diffusion is negligible and the predominant mechanisms for the movement of the aforementioned particles are convection and thermophoresis.[39] For an airborne particle much smaller than the mean free path of air molecule, λ ($\lambda = 0.065$ μm at 298 K and 1 atm), the particle motion is in the so-called free-molecular regime where the Knudsen number $Kn = \lambda/d_p >> 1$. The calculation of the thermophoretic force in this regime has long been acknowledged to be in good agreement with experimental data.[30] However, due to the boundary layer effect and other complexities, the theoretical development of the thermophoresis in the transition regime ($Kn \sim 1$) and the slip-flow regime ($Kn < 1$) is still evolving. The available formulas for these two regimes should be considered as tentative. Talbot et al.[29] have proposed an expression to calculate the thermophoretic velocity applicable to all Knudsen numbers. This formula has been received quite favorably by other researchers[10,39] and is given as

$$v_t = -K\frac{\nu}{T}\nabla T \tag{3.13}$$

and

$$K = 2C_S\frac{\left(k_g/k_p + C_t\lambda/R_p\right)\left[1+\left(\lambda/R_p\right)\left(1.2+0.41e^{-0.88R_p/\lambda}\right)\right]}{\left(1+3C_m\lambda/R_p\right)\left(1+2k_g/k_p+2C_t\lambda/R_p\right)} \tag{3.14}$$

where K is the thermophoretic coefficient, T is the ambient absolute temperature of the gas and ∇T is the temperature gradient, $C_s = 1.147$, $C_m = 1.146$, $C_t = 2.2$, k_g is the thermal conductivity of the surrounding gas, and k_p is the thermal conductivity of the particle. For various k_g/k_p ratios, K in Eq. (3.14) is plotted against λ/R_p as shown in Figure 3.8.[39] Adjacent to a heated planar surface, within the laminar boundary layer, a particle-free zone about one half of the boundary-layer thickness was reported by Talbot et al.[29] However, as particles

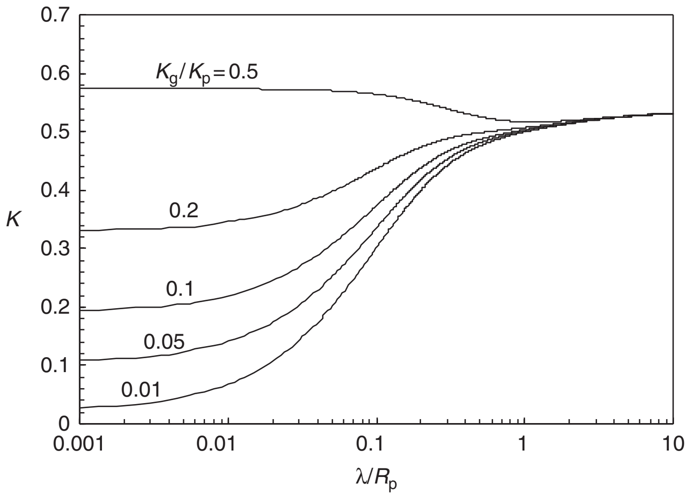

FIGURE 3.8 Values of the thermophoretic coefficient **K** for a spherical particle of radius *a* according to the expression in Eq. (3.14). λ is the mean free path of gas molecules and k_g/k_p is the ratio of the thermal conductivity of the gas to that of the particle.[39]

may rotate in the boundary layer under certain conditions, Chomiak and Gupta[43] and Batchelor[44] argued that the effect of thermophoresis on particle removal might not be as conspicuous as predicted from Eqs. (3.13) and (3.14). As reported in Chomiak and Gupta[43], in a laminar boundary layer, thermophoretic force would not affect the particle if its radius, R_p, is

$$R_p > \left(\frac{18 v D_{T_p} k_g / k_p}{\dfrac{2\lambda^*}{\alpha R_p} + \dfrac{1}{1 + \lambda^*/R_p}} \right)^{1/2} \frac{\mathrm{Re}_x^{1/4}}{U_\infty} \tag{3.15}$$

where D_{Tp} is the thermal diffusivity of the particle material, α is the accommodation coefficient, $\lambda^* = \{(2\lambda C_p/C_v)/[\mathrm{Pr}(C_p/C_v+1)]\}$ is the heat conduction mean free path, C_p and C_v are the specific heats of the gas at constant pressure and volume, respectively, Pr is the Prandtl number, Re_x is the Reynolds number of the flow depending on the distance from the leading edge, and U_∞ is the gas velocity outside of the boundary layer. Likewise, in a turbulent boundary layer, the critical particle size is given as

$$R_p > \left(\frac{202.7 v D_{T_p} k_g / k_p}{\dfrac{2\lambda^*}{\alpha R_p} + \dfrac{1}{1 + \lambda^*/R_p}} \right)^{1/2} \frac{\mathrm{Re}_x^{0.1}}{U_\infty} \tag{3.16}$$

While applying thermophoresis as a cleaning mechanism to remove deposited particles in a naturally convective airflow, as observed in most cleanroom operations, the conditions indicated in Eqs. (3.15) and (3.16) are rarely encountered, as reported by Nazaroff et al.[10,45]

Experimental investigations on thermophoresis have been reported by a number of researchers. Like the corresponding theoretical development in the transition and slip-flow regimes mentioned above, the complexity of the phenomenon has limited the quantity and quality of the experimental data available for model validation. The greatest hindrance to obtain reliable measured data in thermophoresis studies is the interference of natural convection generated also by a temperature gradient in normal gravity.[33,34] By conducting thermophoresis experiments in a microgravity environment, Toda et al. produced some interesting data as listed in Tables 3.4 and 3.5.[33,34] The comparison between the thermophoretic velocity measured by Toda et al.[33,34] and the predictions using Eq. (3.17), derived by Talbot et al.,[29] for Kn < 1,

$$v_t = -2C_S v \frac{\left(k_g/k_p + C_t\lambda/R_p\right)}{\left(1 + C_m\lambda/R_p\right)\left(1 + 2k_g/k_p + 2C_t\lambda/R_p\right)} \frac{\nabla T}{T} \tag{3.17}$$

is shown in Figures 3.9 and 3.10.[29,30,33,34] As shown in Figure 3.9, Eq. (3.17) has substantially under-predicted the thermophoretic velocity for the SiO_2 particles (size ranging from 1 to 30 μm). For the polymethyl methacrylate (PMMA) particles (size ranging from 5 to 30 μm), under-predictions of thermophoretic velocity exist for $d_p < 25$ μm particles and a slight over-prediction for $d_p = 30$ μm particles, as shown in Figure 3.10. The above discrepancies indicated the importance of gravity effect. Hence the existing theories on thermophoresis could be biased due to the fact that they were deduced from experimental data in the presence of gravity.

3.5 ELECTROSTATIC FORCE AND ITS RELEVANCE IN SURFACE CLEANING

As shown in Figure 3.4, an artificial electrostatic field was created to remove particles deposited on microelectronics products.[7,46,47] On the other hand,

TABLE 3.4 Temperature Gradient and Thermophoretic Velocity[33]

Particle <Diameter>	Temperature Gradient ∇T (K/mm)	Thermophoretic Velocity Predicted Using the Theory in[29] v_t (mm/s)	Thermophoretic velocity measured in[33] v_t (mm/s)
MgO	16	0.042	0.22 ± 0.021
<20 μm>	45	0.13	0.67 ± 0.097
SiO_2	22	0.20	0.58 ± 0.017
<20 μm>	80	0.86	3.00 ± 0.094

TABLE 3.5 The Measured Temperature Gradient and Thermophoretic Effect

Particles			
Material	Diameter (μm)	Temperature Gradient ∇T (K/mm)	Particle Velocity Measured in[34] v_t (mm/s)
SiO_2	1.0	40.0	1.68
SiO_2	2.7	22.0	0.58
		80	3.00
		40.0[a]	1.03
SiO_2	10	40.0	0.69
SiO_2	30	40.0	0.53
PMMA	5.0	31.4	0.55
		49.5	1.14
		40.0[a]	0.80
PMMA	12	31.4	0.40
		42.0	0.73
		40.0[a]	0.66
PMMA	30	31.4	0.35
		49.5	0.78
		40.0[a]	0.53

[a] *These values are obtained by interpolation using two preceding data values.*

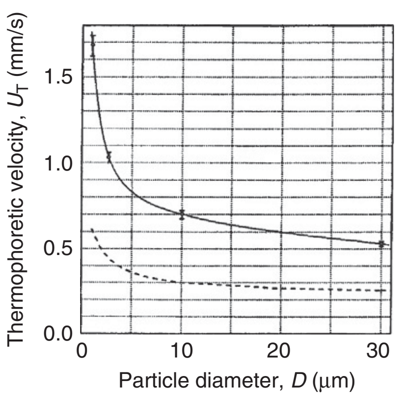

FIGURE 3.9 The relation between the diameter and velocity of the SiO_2 particles ($\nabla T = 40$ K/mm). Solid line: measured in Ref. 34. Dotted line: predicted values based on the Brock equation in.[29,30]

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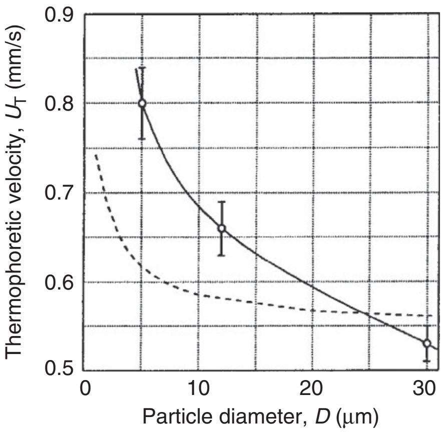

FIGURE 3.10 The relation between the diameter and velocity of the PMMA particles ($\nabla T = 40$ K/mm). Solid line: measured in Ref. 34. Dotted line: predicted values based on the Brock equation in Refs. 29,30.

similar to the thermophoretic effect, the electrostatic force is another mechanism that would affect particle deposition on surfaces.[48–50] Electrostatic phenomena have been observed since the early period of human civilization, but electrostatic force had not been quantified until the discovery of the Coulomb's law that laid the foundation for the science of electrostatics two centuries ago. Even though a primitive electrostatic device for cleaning polluted air was built in 1885, it was never developed to the stage suitable for commercialization.[51] Electrostatic precipitators have been widely used to collect fine particles such as fly ashes from coal-fired furnaces or reactors. Another example of the application of the effect of electrostatic precipitation is the electrostatic air cleaners for household use.

Particles may acquire charges through surface contact, collection of atmospheric electricity, ionization, and similar phenomena.[52] For a conductive particle, the acquisition of charge on its surface is almost instantaneous. The total charge on the surface of a particle, q, is given as

$$q = \varepsilon_0 \pi^3 \mathrm{d}_\mathrm{p}^2 E/6 \tag{3.18}$$

where ε_0 is the dielectric constant of vacuum and E is the electric field strength. For particles with different electrical charges, it was theoretically shown that for dissimilar particles their agglomeration rates were enhanced with increasing surface charge and particle size polydispersity.[53] To remove conductive spherical particles adhered to a conductive surface, the force required to overcome the adhesion force, F_e, is expressed as[46,47]:

$$F_\mathrm{e} = 1.37\pi\varepsilon_0 E^2 d_\mathrm{p}^2 \tag{3.19}$$

For several simple configurations such as parallel plate, tubular, and wire–plate geometry, the electric field distribution could be obtained either by measurement or by solving the governing equation of electrical potential distribution numerically. Figure 3.11 illustrates the relative strength of adhesion, gravitational and electrostatic forces with respect to various particle sizes. To remove particles of various sizes and chemical compositions, the required electric fields are listed in Table 3.6.[46]

3.6 DIELECTROPHORESIS AND ITS RELEVANCE IN SURFACE CLEANING

When a neutral particle of dielectric material is placed in an electric field, the particle becomes polarized and behaves like an electric dipole. If the electric field is not uniform, the force exerted on the particle is not balanced and results

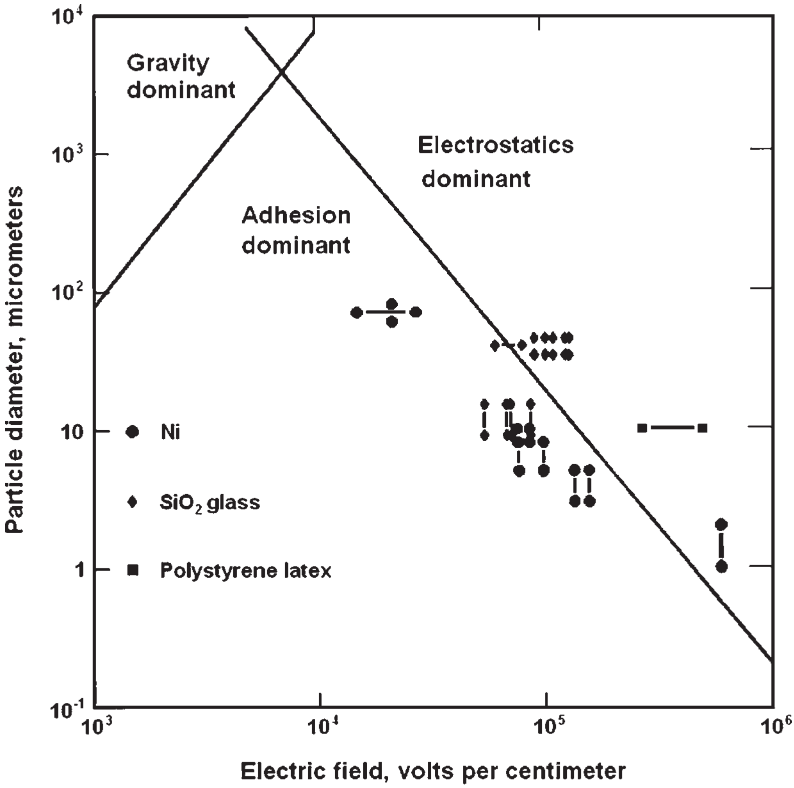

FIGURE 3.11 Particle sizes for which the electrostatic force just equals that of adhesion or of gravitation (lines). The legends shown on the lower left corner on this plot indicate the type of the particle used. "●" are the Ni particles. "◆" are the SiO_2 glass particles. "■" are the polystyrene latex particles.[46]

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TABLE 3.6 **Electric Fields Needed to Remove Particles of the Indicated Size and Composition**[46]

Particle		Field (kV/cm) Needed to Remove the Percentage of Particles Indicated		
Material	**Diameter (μm)**	**5%**	**50%**	**95%**
Glass	34–35[a]	27	72	116
	9–15	20	55	91
		29	69	109
		22	72	123
		20	88	156
SiO_2	0.52	>525		
Nickel	60–75	10	21	32
	8–10	18	87	155
		<0	77	163
	5–8	27	100	174
		12	78	144
	3–5	29	159	289
		46	137	227
	1–3	600[b]		
			600[c]	
Polystyrene	10[d]	41	243	446
		70	367	644
	1.1	>550[e]		

The symbol > means a higher field is needed to remove 5%.
[a]*Data from 11 runs, taken after 60 min on fully charged particles.*
[b]*Two of 35 removed.*
[c]*23 of 43 removed.*
[d]*Only 25 particles total, both runs; not likely fully charged.*
[e]*One of 109 removed; not likely fully charged.*

in the movement of the particle. This phenomenon is called dielectrophoresis.[54–56] In the natural environment, most of the particles are not spherical in shape. Therefore, the effect of particle shape plays an important role in the deposition of non-spherical particles. For instance, much effort has been invested to study the deposition of fibers along the human respiratory tract.[56] It is imperative to sort out particles by their specific aspect ratio or by length for the applications mentioned above. For a conductive fiber with an aspect ratio, $\beta = L/d_p$, its dielectrophoretic velocity is given by Lipowicz and Yeh[56] as

$$v = \lim \beta \to \infty \frac{K_m \varepsilon_0}{24\mu} L^2 \frac{\ln 2\beta - 0.5}{\ln 2\beta - 1} \nabla E^2 \tag{3.20}$$

where v is the dielectrophoretic velocity, K_m is the dielectric constant of the electrically insulating medium, L is the length of the fiber, and E is the strength of the electric field. As indicated in Eq. (3.20), v is proportional to the square of the length of the fiber. Therefore, dielectrophoresis is a more effective fiber classifier than electrophoresis, as recommended by Lipowicz and Yeh.[56] In the presence of a strong electric field, to avoid the interference by electrophoresis, it is necessary to use either uncharged particles or use an alternating (AC) electric field to fully utilize the effect of dielectrophoresis.[55,56]

3.7 ABRASIVE EROSION AND ITS RELEVANCE IN SURFACE CLEANING

Solid particle erosion or abrasive blasting erosion refers to the general mechanical degradation or wear of a solid material subjected to a jet stream of abrasive particles impinging on its surface. Examples of the application of abrasive erosion can be found in abrasive cleaning of metal surfaces for coating[58] and for stress corrosion cracking investigation,[59] in chemical mechanical polishing,[60] and in abrasive drilling and cutting by water jet.[61,62] However, in many other applications, abrasive erosion causes sever surface damage, which is, of course, not desired. Abrasive erosion damage has been extensively reported, such as in rocket motor tail nozzle,[63] on helicopter rotors and gas turbine blades,[64] and in pipeline systems for pneumatic or hydraulic transportation of solids.[4,65] In this section, some recent developments on fundamental mechanisms of solid particle erosion, including abrasive erosion by a single particle impact and erosion by multiple impacts of a solid particle stream, as well as parametric effects on the abrasive erosion rate are discussed.

When a solid particle impinges on a material surface, it may skid and/or rotate on the surface and at the same time create an indentation (ductile materials) or crater (brittle materials). With successive impacts of particles on the same location, this indentation causes material removal due to the repeated plastic deformation and the skidding of sharp-edged hard particles over a surface of relatively low hardness. When the impact velocity is lower than the critical impact velocity, the particle does not skid but rotate, resulting in no cutting damage but only plastic fatigue damage. When the impact velocity is higher than the critical velocity, the solid particle will skid on the surface, creating a crater, and then rotate on the surface, as shown in Figure 3.12. The critical impact velocity, V_c, can be estimated based on a rolling/sliding contact mechanism,[66] as:

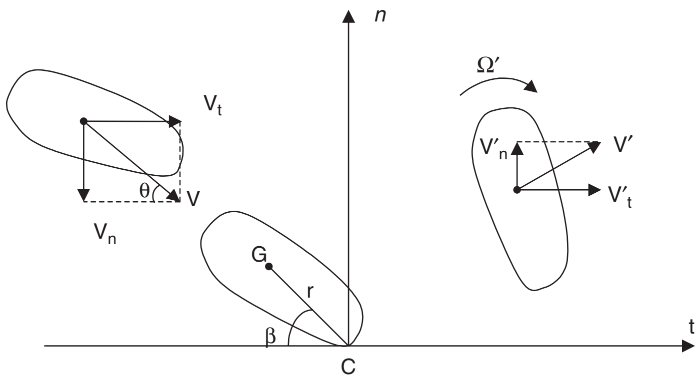

FIGURE 3.12 Schematic diagram of surface impact of an abrasive particle.

$$V_c = \frac{K + \cos\beta(\cos\beta - f\sin\beta)}{f(K + \sin\beta) - \cos\beta} \frac{k_1 d}{(1+e)\left(\pi\sqrt{\dfrac{m}{2\pi p r_s}} - k_e V_n^{-1/5}\right)} \frac{\sigma_y}{10E\sin\theta} \tag{3.21}$$

where θ is the impact angle, β is the contact angle, V_n is the normal component of impact velocity, p is the yield pressure, r_s is the equivalent radius of a solid particle, d is the maximum crater diameter, m is the particle mass, k_e and k_1 are material property related constants, f is the friction coefficient, E is elasticity or Young's modulus, σ_y is the yield stress, e is the rebounding coefficient, and K is the shape factor of particle. The friction coefficient was further estimated by Yabuki and Matsumura[66] and given as:

$$f = \frac{\dfrac{\pi}{8}d^2 s + \dfrac{1}{12}\dfrac{d^3}{r_s}p'}{m(1+e)V_n} \times \left(\frac{\pi}{2}\sqrt{\frac{m}{2\pi p r_s}} + \frac{k_e}{2}V_n^{-1/5}\right) \tag{3.22}$$

where p' is the average pressure required to displace the soft material from the front of the particle and s is the shear strength of the soft material. The rebounding coefficient was calculated by Tabor[67] using a normal impact model and is given by

$$e = \frac{\left(\dfrac{10^4}{\pi^5 4^3 3^5}\dfrac{m}{2r_s^3}\right)^{-1/8}\left(\dfrac{1-v_p^2}{E_p} + \dfrac{1-v_s^2}{E_s}\right)p^{5/8}}{V_n}\left(V_n^2 - \frac{3}{8}V'^2_n\right)^{3/8} \tag{3.23}$$

where e is the restitution coefficient (or rebounding coefficient), v is the Poisson's ratio, and V'_n is the normal component of rebounding velocity.

The most important issue in abrasive erosion is the erosion rate. The volume of material, Q, removed by a single abrasive impact was given by Finnie[68] and is expressed as:

$$Q = \begin{cases} \dfrac{mV^2}{\rho\psi k}\left(\sin 2\theta - \dfrac{6}{k}\sin\theta\right) & \tan\theta \le \dfrac{k}{6} \\ \dfrac{mV^2}{\rho\psi k}\left(\dfrac{k}{6}(\cos\theta)^2\right) & \tan\theta > \dfrac{k}{6} \end{cases} \tag{3.24}$$

where ψ and k are the ratio of length to depth of the scratch formed and the ratio of vertical to horizontal force components, respectively, ρ is the energy required to remove a unit volume of material from the body. Winter and Hutchings[69] further identified two mechanisms for volume removal, namely, ploughing and cutting. Ploughing occurs when a particle rolls over, instead of sliding along the surface, which causes the cutting edge of the particle to penetrate deeply into the material surface instead of performing a scooping action. The total volume removed by an abrasive impact comes from two contributions: the ploughing loss,[65]

$$Q_{\mathrm{P}} = \frac{0.5m(V\sin\theta - k)^2}{\Omega} \tag{3.25}$$

and the cutting loss,[70]

$$Q_{\mathrm{C}} = \frac{m}{2\phi}\left(V^2\cos^2\theta - V_{\mathrm{r}}^2\right) \tag{3.26}$$

where Ω and ϕ are the energies required to remove a unit volume of material from the body by ploughing and cutting, respectively, k is a material property related constant, and V_{r} is the particle rebounding velocity.

Despite a relatively clear understanding of abrasive erosion by a single particle impingement, the current knowledge on abrasive erosion by a jet stream of abrasive particles is still inadequate to provide predictive and reliable information on the erosion rate for a given engineering design or application. This lack of knowledge has resulted from insufficient information in three interrelated areas: (1) definition of erosion resistance as an objective material property; (2) establishment of the quantitative relationship among erosion rate, erosion resistance, and environmental parameters; and (3) the determination of these environmental parameters for a given application. So far most studies on erosion rate are reported in terms of the material loss curves or loss rates for particular types of materials of interest. The erosion rate is defined as the ratio of mass loss to mass of erodent particle, which is a dimensionless quantity.

There have been many attempts to correlate the erosion rates measured for different materials with one or more established mechanical properties of these materials.[71–73] A theoretical dynamic analysis of a new laboratory technique was recently proposed by Talia et al.[74] for investigating the abrasive erosion mechanisms of brittle, ductile and semi-ductile materials. As shown in Figure 3.13, the solid material samples are mounted on a rotating disk, which are air-blasted by abrasive particles. This measurement system is easy to individually investigate the effects of abrasive impact angle, impact velocity components (normal and tangential), and abrasive material properties. Typical effect of the impact angle on erosion rate is shown in Figure 3.14, in which a nonlinear relationship between erosion rate and impact angle is seen. In addition, the erosion loss appears to be undetectable at small impact angles, which indicates that a minimum angle is required for material removal. Typical effect of impact velocity on erosion rate is illustrated in Figure 3.15, in which

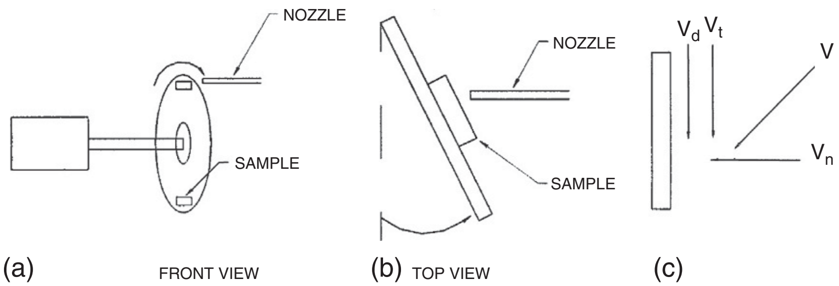

FIGURE 3.13 Experimental system for study of particle erosion mechanism.[74]

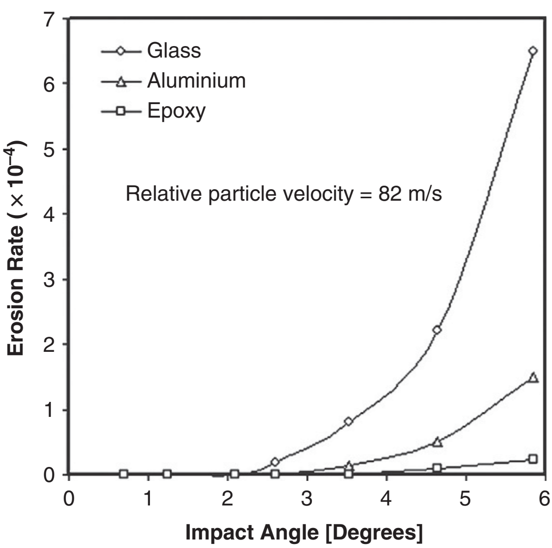

FIGURE 3.14 Example of effect of impact angle on erosion rate.[74]

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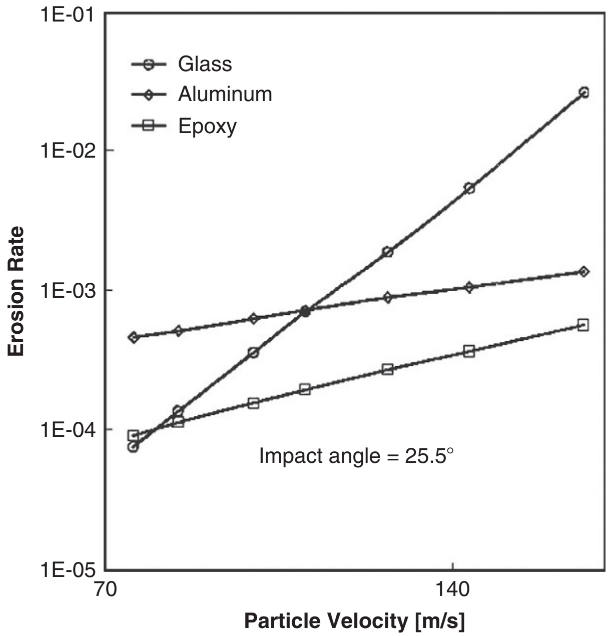

FIGURE 3.15 Example of effect of normal impact particle velocity on erosion rate.[74]

the erosion rate of different materials representing ductile, brittle and semi-ductile at various impact velocities but with the same impact angle are presented. The apparent linear relationships between the erosion rate and normal impact velocity in a semi-logarithm scale indicate certain similarities of the materials tested.

3.8 SUMMARY

Particle deposition problems are vital challenges to the practitioners in the industries dealing with contaminant removal and surface cleaning. In many manufacturing and processing environments, such as that for producing semiconductors and integrated circuits, substantial amounts of time and efforts are devoted to tackle the ubiquitous problems of particle deposition. Multiple mechanisms are involved in the deposition of particles on various surfaces encountered in industrial processes. In this chapter, the most common and influencing mechanisms of particle deposition are described. The physical aspects of the effect due to dry deposition, thermophoresis, electrostatic force and dielectrophoresis on particle deposition on surfaces are delineated and emphasized to strengthen their relevance in contaminant removal and surface cleaning. The applicable range of a specific theory and related caveats when applied to an industrial setting are also provided throughout this chapter. For an in-depth understanding of each of the mechanisms involved in particle deposition, the references cited in this chapter should serve as further reading material and guide to the specific knowledge as well.

Surface cleaning by mechanical means can be achieved by abrasive erosion from impingement of fine and hard particles on the surface to be cleaned. The effectiveness of abrasive erosion depends on impact velocity, impact angle, and materials involved. The available empirical correlations are typically application oriented, namely, they are valid for specific applications and within a narrow range of parametric conditions. One particular difficulty in model generalization of abrasive erosion for practical applications is the low predictability of particle behaviors in turbulent jets and boundary layer flows over rugged (eroded) surfaces. Theories with a much wider applicability and a deeper scientific basis are yet to be developed.

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